BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
Department of Computer Science and Engineering

Script Matters

Measuring Latin-Script **Banglish Robustness** in Bangla LLMs

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Motivation | One question, more than one way to type it

Bangla is normally written in **Bengali script**. But in everyday digital life—messaging, search, comments, reviews—many people type Bangla with **Latin letters**. This form is called **Banglish**.

For a Bangla-facing LLM, the *same user intent* can arrive through more than one written form—and the model may answer one and fail another, **even though the gold answer never changed**.

270M+

Bangla speakers world-wide—7th most-spoken language

অ → a

Banglish needs only a Latin keyboard—the path of least resistance

Bangla

গাছ কীভাবে খাদ্য তৈরি করে?

Banglish

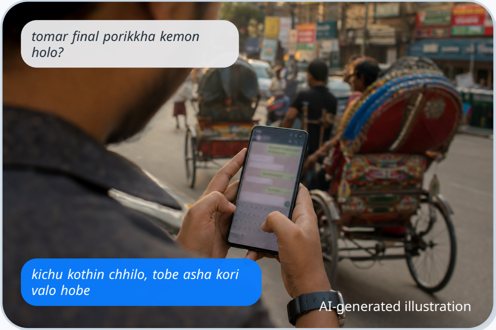
gach kibhabe khaddo toiri kore?

English

How do plants make food?

*Same item id, same gold answer. Only the **script** changes.*

Banglish in the Wild | This is how millions already type Bangla



AI-generated illustration

dengue jor er lokkhon ki ki

dengue jor **koto din thake**

dengue hole **ki khabar khete hobe**

About 2,40,000 results (0.42 seconds)

★★★★☆ **4.0** **Verified purchase**

product ta darun! delivery onek fast chhilo, packaging o bhalo. dam onujayi worth it.

Helpful (132) • Reply

Millions already type Bangla this way—**does the model stay as reliable when the input is Banglish?**

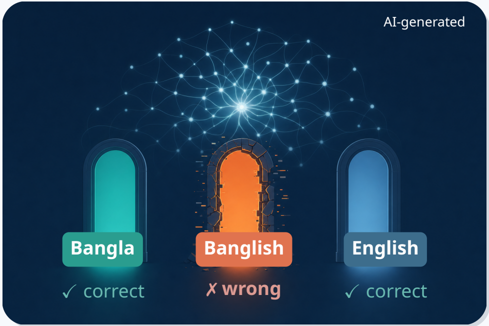
The Hidden Access Gap | Script as a controlled, measurable variable

If the task and gold answer are held fixed, does changing the script to Banglish change model reliability?

This is *not* "is Bangla hard for LLMs?"—already known. The missing piece is narrower:

- **Script**, not language, is the variable.
- Same item, same gold answer—*did it flip?*
- The gap is **measured**, not assumed.

Same user, three doors to the model



Framing: Banglish as an Orthographic Access Problem

Central claim. Script choice is a *measurable robustness variable*: it can change accuracy, strict answer compliance, recoverability, and cost under controlled conditions.

Three design principles guide the whole study:

1. Paired

Same item id & gold answer across all three scripts. Did *this* item flip?

2. Audited

Banglish is *human-reviewed*, not raw machine transliteration. Rules out "bad Banglish."

3. Boundary

Stronger models are a *boundary test*, not a shortcut. Where does the gap end?

Not that Banglish is always harder—that reliability must be measured directly, not assumed.

Related Work | Three strands of prior work—and what each leaves open

Bengali task evaluation

BEnQA (Shafayat et al., 2024) • **BanglaMATH** (Prama et al., 2025) • BnMMLU, BLUCK, BanglaQuAD, NCTB-QA ...

Active landscape—but script is never the controlled variable.

Romanized & code-mixed Bangla

BanglaTLit (Fahim et al., 2024) • BanglishRev • BanTH • BnSentMix

Prevalence evidence—classification corpora, not paired QA.

Script robustness—nearest neighbors

Translit. perturbations (Haider et al., 2025) • Script Gap (Khullar et al., 2025) • Romanized Nepali (Rimal & Rimal, 2026)

No gold-answer QA/math; no reviewed romanization.

Tokenization & latent language

Tokenizer unfairness (Petrov et al., 2023) • latent English (Wendler et al., 2024) • RomanLens (2025)

Motivates our token & oracle analyses—doesn't predict robustness.

None of these measures whether the **same item, same gold answer** gets harder when only the **script** changes.

Where We Sit | Against the nearest neighbors, feature by feature

Each nearest neighbor has one or two of the needed design elements—none has them together.

Work	Paired items	Human-rev. roman.	Downstream QA/math	Mitig. study	Scale ext.
Translit. perturbations (Haider et al., 2025)	◐	✗	✗	✗	✗
Script-gap triage (Khullar et al., 2025)	✓	✗	✗	✗	✗
Romanized Nepali (Rimal & Rimal, 2026)	◐	✗	✗	✓	✗
This thesis	✓	✓	✓	✓	✓

✓ covered ◐ partial ✗ absent

We combine **all five** for the first time, under one protocol—the first paired, human-reviewed, multi-model Banglish robustness study with downstream QA/math and a scale extension.

Contributions | Five, under one protocol

1. Paired BN / BG / EN protocol

Same QA & math items in three script views; item id, answer type, and gold answer preserved.

2. Reviewed gold-core set

200-item slice (144 BEnQA + 56 BanglaMATH); Banglish *frozen* after targeted human review.

3. Not cleanup or token length

Review barely moves scores; Banglish is *token-cheaper* than Bangla yet scores lower.

4. Frontier & scale boundary

5 hosted models + a **974-row gold extension** (1,174 paired items). Every model—incl. GPT-5.5—keeps a significant deficit at scale.

5. Negative mitigation evidence

Self-normalization, generated views, and LoRA all fail to give a general fix—the weakness runs deep.

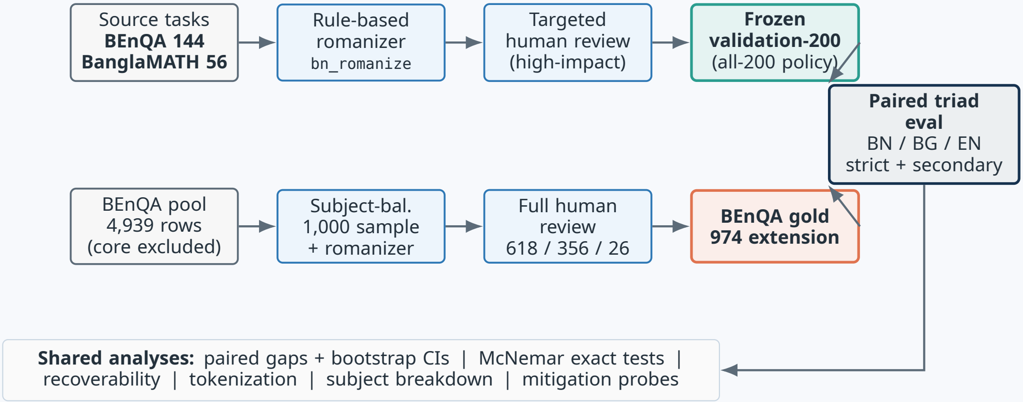
One-line thesis: Banglish is an **undermeasured orthographic robustness challenge**—measure it, don't assume it.

SECTION 2 / 6

Benchmark & Protocol

How we make *script* the only variable

Benchmark Construction | Two tracks, one paired protocol



Validation-200 (core): reviewed, mixed-task, frozen *before* seeing outcomes.

974 extension: BEnQA-only, fully human-reviewed, for scale.

Evaluation Protocol | Models, scoring, and the paired gap

Open Qwen triad — reproducible baselines

Qwen2.5-3B | Qwen2.5-7B (8-bit) | Qwen3-4B

Frontier / API panel — boundary test

Gemini 3.5 Flash | GPT-5.5 low | Claude Sonnet 4.6 | DeepSeek V4 Flash | Llama 3.3 70B (Groq)

8 models $\times$ 3 script views $\times$ paired items

Scoring — two lenses

Strict — deployable form, the contract.

Secondary — credit for recoverable answers (units, light markdown).

The paired gap (per model, per item):

$$\Delta = \text{acc(reviewed Banglish)} - \text{acc(Bangla)}$$

Negative $\Rightarrow$ Banglish less accurate.
Paired bootstrap CIs + Holm-corrected **McNemar exact** tests.

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Main Results

Reviewed Banglish is consistently harder

Main Result | Validation-200, strict accuracy

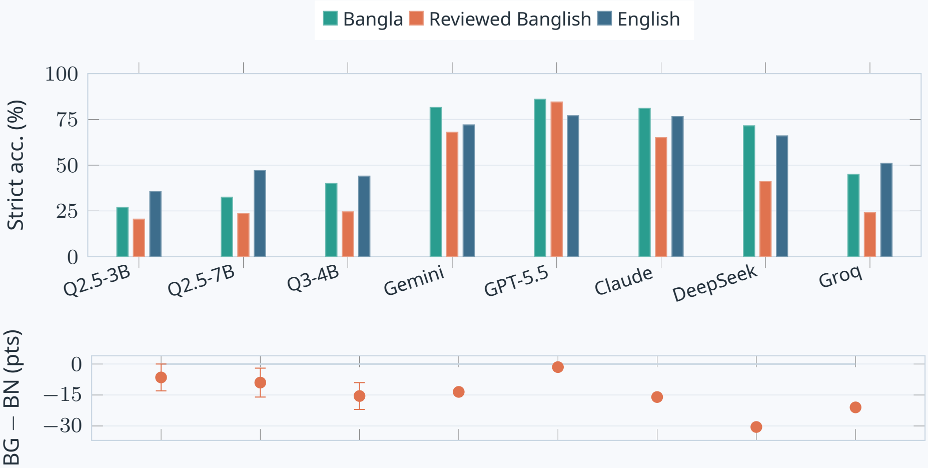
Model	Bangla	Rev. Banglish	English	BG − BN	BG − EN
Qwen2.5-3B	54	41	71	−6.5	−15.0
Qwen2.5-7B 8-bit	65	47	94	−9.0	−23.5
Qwen3-4B	80	49	88	−15.5	−19.5

bars: items correct out of 200, strict scoring

For **every** Qwen model, reviewed Banglish is the **lowest** script view—below both native Bangla *and* English.

- BG − BN: −**6.5 to** −**15.5** pts
- BG − EN: −**15 to** −**23.5** pts — many failures are *not* item difficulty: the same item is answerable in another view.

Main Result | The pattern is visual and consistent



Lowest bar for every model **except GPT-5.5**; the BG−BN interval excludes zero for **7 of 8** models.

Is It Statistically Real? | McNemar exact tests on discordant pairs

The paired design is the textbook McNemar setting: only items where exactly one script is correct count. *b* = Bangla-only correct, *c* = Banglish-only correct.

Model	*b*	*c*	**Odds ratio [95% CI]**	**Exact** *p*	**Holm** *p*
Qwen2.5-3B	28	15	1.84 [0.99, 3.41]	0.066	0.066
Qwen2.5-7B 8-bit	37	19	1.92 [1.11, 3.32]	**0.022**	**0.044**
Qwen3-4B	39	8	4.65 [2.22, 9.75]	$<$**0.0001**	$<$**0.0001**

Qwen3-4B and Qwen2.5-7B stay significant after Holm correction. Qwen2.5-3B is marginal on the 200-core ($p = 0.066$)—but on the **974-row extension** the same model becomes significant ($p = 0.024$). The marginal result reflects *sample size*, not absence of effect.

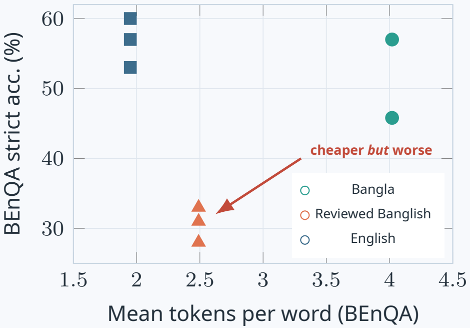
The Gap Survives Every Robustness Check

Could the gap be an artifact? We attacked it from five directions. It survived all of them.

Objection	Test	Outcome
“Banglish is just bad”	v5 targeted human review, then freeze	**$\leq$ 2 items**
“A few bad source rows”	strict-197 (drop 3 flagged rows)	**still -7 to -16**
“It’s the prompt wording”	2 neutral templates (terse, exam-role)	**-3 to -4.5**
“It’s greedy decoding”	temperature 0.7	**-4.5**
“It’s tokenization length”	audited tokens-per-word	**cheaper, not longer**

Review *improves dataset quality* while preserving the central pattern. The gap stays negative under every template and temperature—the weakness is **script-conditioned**, not a wording or decoding accident.

Tokenization | Cheaper but worse—cost and accuracy decouple



Mean tokens / word (audited Qwen tokenizers):

	BN	BG	EN
BEnQA	4.02	**2.49**	1.95
BanglaMATH	4.63	**2.11**	1.41

Reviewed Banglish uses **fewer** tokens per word than Bangla, yet scores **lower** for every model. It rules out the simple “Banglish fails because it is longer” story.

The failure is more consistent with script-conditioned lexical grounding than with context budget.

Spelling-Variation Robustness | Not an artifact of one romanizer

The clearest limitation of a rule-based romanizer: it is not natural, spelling-variable Banglish. So we generated **4 extra seeded respellings** per item (e.g. sh → s, bh → v) for 100 BEnQA items—5 spellings each.

Model	Accuracy	Spelling range	Items flipping	Flip rate
Qwen2.5-3B	31.2%	28–34%	17/100	**17%**
Qwen3-4B	29.0%	28–31%	6/100	**6%**

Magnitude is stable

Accuracy stays in a narrow **28–34%** band across independent spellings—the penalty is *not* one romanizer's quirk.

Items are locally unstable

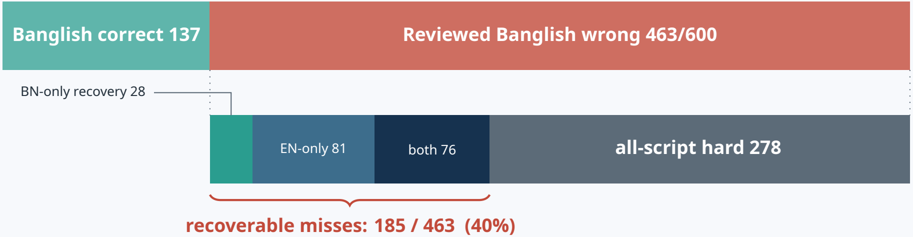
Yet **17%** of items flip correctness when only the spelling changes—same question, same answer.

Why Does It Fail?

Recoverability and error taxonomy

Failure Analysis | A large share of Banglish errors are recoverable

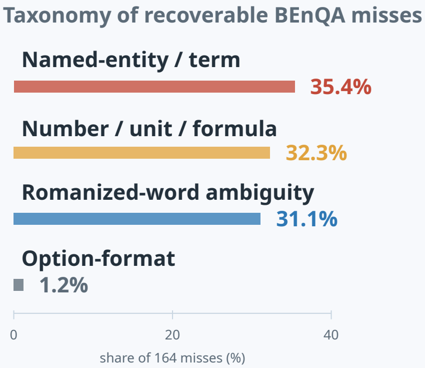
Paired item ids let us separate *hard items* from *script-specific failures*. Across 3 Qwen models $\times$ 200 items = 600 slots:



Yet **not absolute**: 20/600 slots are *Banglish-only* wins. The claim is aggregate and paired—*systematically* less reliable, not always worse.

55% of BEnQA Banglish misses are recoverable: the same answer is reachable in Bangla or English.

Why Banglish Fails | Error taxonomy of 164 recoverable misses



No single dominant failure mode. A narrow parser, prompt, or spelling cleanup cannot close the gap by itself.

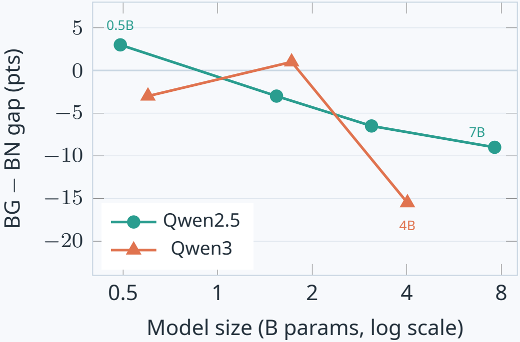
- **Technical terms become opaque:** *lyakotik esid* $\rightarrow$ lactic acid.
- **Numbers, units, formulas drift** in mixed numeric + romanized context.
- **Latin spellings create ambiguity** for Bangla words and named entities.

Format is not the story: option-format failures are only 1.2%.

Frontier & Scale

Does capability or size fix it?

Within-Family Scaling | The gap *widens* with model size



Qwen2.5 shows a **clean monotonic trend**: the gap moves $+3.0 \to -3.0 \to -6.5 \to -9.0$ as size grows, with McNemar p falling $0.26 \to 0.02$.

Qwen3's largest model has **by far the biggest gap** (-15.5, $p < 0.0001$).

> Added capability is preferentially spent on the higher-resource Bangla and English views. The script gap is **not something scaling alone fixes**—within these families it grows.

Frontier & Scale Boundary | The strongest result

GPT-5.5 *nearly closes* the gap on the 200-core (secondary scoring). But scale tells the real story: a 974-row human-reviewed gold extension.

Model (974-row)	**BN**	**Banglish**	**EN**	**BG−BN**	**95% CI**	**McN. p**
Qwen2.5-3B	33.2	29.3	50.3	**−3.9**	[−7.2, −0.5]	0.024
Gemini 3.5 Flash	76.3	65.0	69.8	**−11.3**	[−13.7, −8.9]	$<.0001$
GPT-5.5 (none)	84.2	71.8	84.7	**−12.4**	[−15.1, −9.8]	$<.0001$
Claude Sonnet 4.6	78.4	53.8	79.2	**−25.4**	[−28.6, −21.9]	$<.0001$
DeepSeek V4 Flash	77.6	45.0	81.2	**−32.7**	[−36.0, −29.3]	$<.0001$
Groq Llama 3.3 70B	56.2	34.2	63.9	**−22.0**	[−25.7, −18.2]	$<.0001$

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models keep a significant deficit at scale

−12.4
GPT-5.5 still fails (pts, $p < .0001$)

−32.7
DeepSeek—worst, despite high BN/EN

1,174
paired items (200 core + 974 gold)

GPT-5.5's near-closure on 200 items does *not* generalize—**high absolute accuracy ≠ Banglish robustness.**

SECTION 6 / 6

Mitigation

What if we try to fix it?

Mitigation Attempts | An honest negative result

We tried the natural fixes on *both* sides of the model. None gives a general, deployable solution.

Approach	Finding	Verdict
Self-normalization	Helps Qwen2.5-3B (+6.5) but *hurts* Qwen3-4B (−12.5)	**model-dependent**
Generated alt-script views	Routing too sparse; views not reliable enough	**brittle**
LoRA adapter (eval-disjoint)	Gap $-6.5 \to -5.5$ / -4.0; *not* significant	**slight,** $p > 0.3$

Light, eval-disjoint LoRA—the natural first training-side fix—narrows the gap only slightly and **not reliably**. The weakness is **deeper than a small adapter can close**. *A negative result is still useful.*

AI-generated illustration

Conclusion

What we showed

- ✓ Script choice **changes reliability** under controlled conditions—surviving review, tokenization, scaling, and frontier comparison.
- ✓ The gap is **statistically significant** for every model at scale, including GPT-5.5.
- ✓ High absolute accuracy does **not** guarantee Banglish robustness.
- ✓ Naive mitigation is **brittle**; the weakness runs deep.

Why it matters

The practical lesson

For Bangla users, Latin-script input is a **real access path**—not informal noise models will automatically handle.

Robust evaluation must **measure it directly**; robust mitigation must **preserve task, format, and script equivalence**.

Banglish is an undermeasured orthographic robustness challenge for Bangla LLM use.

Questions & discussion

Key takeaway: Banglish robustness must be measured directly, not inferred from Bangla or English.

AI-generated illustration

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